

Direct vs Edge Matrix

Fifteen criteria are compared; no topology winner is declared.

CO-APPLICABLE / NOT SEQUENTIAL / NOT SELECTED

CRITERION		DIRECT CONNECTIVITY PLAUSIBLE ALTERNATIVE	EDGE AGGREGATION PLAUSIBLE ALTERNATIVE
01 Operator-facing connectivity identity count		HIGH — one per device	LOW — one per aggregation group
02 Device independence		HIGH — each device operates without dependency on another local device	LOW — devices depend on the shared aggregation function's availability
03 Local power dependency		LOW — only the device itself needs power	MEDIUM–HIGH — aggregation function also needs power; exact value TBD pending site survey
04 Buffering		Per-device (distributed)	Centralized at aggregation function or still per-device — TBD
05 Aggregation		Not applicable	Native to this pattern
06 Physical distribution		Devices spread across building without a common local node	Devices cluster around a common local node
07 Maintenance complexity		MEDIUM — more individual connectivity endpoints to manage	MEDIUM — fewer connectivity endpoints, one added local component to maintain
08 Lifecycle complexity		MEDIUM — device lifecycle only	MEDIUM–HIGH — device plus aggregation-function lifecycle
09 Failure-domain size		LOW per device — isolated failure	HIGHER for group — aggregation-function failure affects all devices behind it RELATED: A3-EC03-06 / EC-03 §29 F-03
10 Single-point-of-failure exposure		LOW	HIGHER — aggregation function is a candidate single point of failure for its group
11 Local protocol dependency		NONE — no local link required	TBD — unspecified local link required between devices and aggregation function
12 Operator visibility		Per-device connectivity state	Per-aggregation-group connectivity state; device-level detail potentially reduced
13 Site-survey dependency		Per-device location only	Per-device location plus aggregation-function location and local-link feasibility
14 Scalability		MEDIUM — each additional device adds its own connectivity identity	HIGH within cluster — devices can join existing aggregation function without new external connectivity identities
15 Replacement impact		LOW — replacing one device does not affect others	MEDIUM — replacing aggregation function could affect all devices behind it; replacing one device does not

OPEN / TBD — 4 ITEMS

EC03-04-TBD-01 Edge local-power dependency exact value; site survey required.
EC03-04-TBD-02 Edge buffering placement; joint device/integration validation required.
EC03-04-TBD-03 Edge local-link dependency / site feasibility; joint validation required, including Beeline where its boundary is affected.
EC03-04-TBD-04 Direct versus Edge selection = NOT SELECTED; later decision gate required.

RATING VOCABULARY LOW / MEDIUM / HIGH / TBD
Source-preserved qualifiers: MEDIUM–HIGH / HIGHER / NONE / Not applicable. Qualifiers and reasoning are not a calculated score or severity heatmap.

RESPONSIBILITY CONTEXT
Beeline — CONNECTIVITY / TRANSPORT PARTNER
HouseMaster — INDEPENDENT DOMAIN LAYER / SYSTEM OF RECORD
Trust/API Boundary NOT SHOWN. No EDGE-01 instantiation. No deployment inference.

NO TOPOLOGY WINNER DECLARED • NO TOPOLOGY RECOMMENDED OR SELECTED • NO COST FIGURE USED • ARCHITECTURAL OPTION ≠ DEPLOYMENT DECISION

REFERENCE SEM-MCD-001 — Semey, Shakarima 13A | Installed devices: 0 CONFIRMED | Actual installed-device presence: NOT ESTABLISHED | 0 CONFIRMED ≠ PROVED ABSENCE